United States Patent Application Publication	20250281959
Kind Code	A1
Publication Date	September 11, 2025
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MANUAL PIPE BENDING APPARATUS HAVING IMPROVED BENDING WORK CONVENIENCE

Abstract

Provided is a manual pipe bending apparatus having improved bending work convenience, the apparatus having an external gear formed on an outer peripheral surface of a bending shoe and having an operating ratchet engaging therewith, so that, when a worker rotates an operating lever to bend a pipe, rotation work, of the worker, for pushing or raising the operating lever is convenient and, when the operating lever is pushed once, the bending shoe can be rotated at a predetermined angle, thereby enabling the pipe to be bent uniformly at a predetermined degree, and thus easy bending work and a uniform bending angle can be expected. Between the operating parts, the operating ratchet which is rotated by a third rotating shaft and which has an end portion engaging with the external gear and an elastic means for applying elastic force to the operating ratchet are provided.

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Family ID: **1000008628486**

Appl. No.: **18/289742**

Filed (or PCT Filed): **March 16, 2022**

PCT No.: **PCT/KR2022/003662**

Foreign Application Priority Data

KR	10-2021-0061964	May. 13, 2021
KR	10-2021-0067359	May. 26, 2021

Publication Classification

Int. Cl.: **B21D7/04** (20060101)

U.S. Cl.:

CPC **B21D7/04** (20130101);

Background/Summary

DESCRIPTION
Technical Field
[0001] The present disclosure relates to a pipe bending apparatus operated in a manual type. More particularly, the present disclosure relates to a manual pipe bending apparatus having improved bending work convenience, the manual pipe bending apparatus being configured such that a worker can easily push and raise an operating lever in bending a pipe by using an external gear formed on an outer peripheral surface of a bending shoe and an operating ratchet engaging with the external gear while the worker rotates the operating lever. Furthermore, due to this structure, the manual pipe bending apparatus of the present disclosure has excellent bending workability compared to bending workability of existing pipe bending apparatuses and is capable of being manufactured in a small and lightweight type, so that the manual pipe bending apparatus has improved mobility and is capable of being carried in a vehicle, thereby ultimately improving the bending work convenience.
Background Art
[0002] Generally, pipes are used for guiding electrical wires or used for transporting fluids (gas or liquid), and there are various types and uses of pipes. Pipes generally require bending in which the pipe is bent at a predetermined angle according to an installation location, and the pipe is used by bending the pipe at various angles such as 90 degrees, 45 degrees, and so on by using a bending apparatus.
[0003] Meanwhile, in the bending apparatus for bending the pipe, as a rotating shaft is gradually driven while the pipe that is the work object is seated and fixed on a bending shoe mounted on the rotating shaft, a rotation of the bending shoe is realized, and the bending of the pipe is performed according to a rotation angle of the bending shoe.
[0004] In addition, the configuration of the bending apparatus varies according to whether the bending shoe is rotated automatically or manually, and the present disclosure relates to a manual pipe bending apparatus in which a worker directly moves a lever up and down so as to operate a bending shoe.
[0005] Such a manual pipe bending apparatus is mainly used for bending of a small pipe compared to an automatic pipe bending apparatus, and the bending of the pipe in the manual pipe bending apparatus is performed on the basis of the worker's sense of rotating the lever, i.e., the bending of the pipe in the manual pipe bending apparatus is performed on the basis of the worker's experience of how much force the worker pushes the lever so as to rotate the bending shoe when the worker pushes the lever once.
[0006] As a manual pipe bending apparatus, Korean Patent No. 10-1861444 (May 18, 2018) filed and registered by the applicant of the present disclosure is provided.
[0007] In Korean Patent No. 10-1861444, a manual pipe bending apparatus having a simple structure for fastening components to each other is provided, but a rotation angle of a bending shoe when a worker rotates a lever once is not constant and the worker has difficulty in pushing the lever once smoothly, so that there is a limitation in that workability of the manual pipe bending apparatus in Korean Patent No. 10-1861444 is less than workability of the present disclosure described below.
[0008] Of course, in addition to Korean Patent No. 10-1861444, existing manual pipe bending apparatuses for bending a small pipe have a problem or a limitation mentioned above.

[0009] Furthermore, there are problems that the existing manual pipe bending apparatuses have sizes and weights difficult to be carried in a vehicle and mobilities of the existing manual pipe bending apparatuses are significantly reduced.

DISCLOSURE

Technical Problem

[0010] Accordingly, the present disclosure has been made keeping in mind the above problems and limitations occurring in a manual pipe bending apparatus.

[0011] First, an objective of the present disclosure is to provide a manual pipe bending apparatus configured to facilitate a worker to perform pushing work of a lever for rotating a bending shoe and configured to rotate the bending shoe by a predetermined angle when the pushing work is performed, thereby increasing workability.

[0012] Second, another objective of the present disclosure is to provide a manual pipe bending apparatus capable of realizing the objective described above, the manual pipe bending apparatus also having a simple internal structure and being capable of being manufactured in a small and lightweight type so that a worker can easily move the manual pipe bending apparatus, and the manual pipe bending apparatus being capable of being carried in a vehicle.

Technical Solution

[0013] In order to achieve the objectives described above, according to the present disclosure, there is provided a manual pipe bending apparatus.

[0014] The manual pipe bending apparatus includes: a support positioned on a floor; a support bracket coupled to a first side of the support; a rotating bracket coupled to a first side of the support bracket such that the rotating bracket is capable of being rotated, the rotating bracket having first and second rotating bracket plates which are integrally formed by being connected to each other with a space formed therebetween; a bending shoe coupled to the rotating bracket such that the bending shoe is capable of being rotated between the first and second rotating bracket plates by a first rotating shaft, the bending shoe having an outer peripheral surface provided with a guide groove to which a pipe that is inserted is guided; an operating part positioned above the bending shoe and coupled to the rotating bracket such that operating part is capable of being rotated by a second rotating shaft; and an operating lever coupled to a first side of the operating part and configured to control whether the operating part is rotated or not.

[0015] The guide groove may be formed on a part of the outer peripheral surface of the bending shoe, and an external gear may be formed on the outer peripheral surface of the bending shoe where the guide groove is not formed.

[0016] The working part may have first and second operating plates that are integrally formed by being connected to each other by a second connection pin with a space formed therebetween, and an operating ratchet which is configured to be rotated by a third rotating shaft and which has an end portion engaging with the external gear of the bending shoe and an elastic means configured to provide an elastic force to the operating ratchet may be provided between the first and second operating plates.

[0017] In addition, the external gear according to the present disclosure may be formed such that a center angle between one gear and a next gear of the external gear is 15 degrees.

[0018] In addition, the bending shoe according to the present disclosure may include a mounting jaw which is formed on a first side of the guide groove and on which an end portion of the inserted pipe is mounted, and an incision part in which a part thereof is cut-off may be formed on an upper side of the guide groove where the mounting jaw is positioned.

Advantageous Effects

[0019] According to the configuration described above, there are effects as follows.

[0020] Each time a worker pushes the operating lever once, the operating ratchet is engaging with the external gear, and the bending shoe is rotated by a predetermined angle. Furthermore, when the worker raises the operating lever, an engaged state of the external gear is released and the external gear is in ready for being in the engaged state again. Therefore, the worker can easily operate the operating lever and also a bending angle of the pipe is capable of being uniformly controlled, so that there is an effect that bending workability is increased compared to exiting pipe bending apparatuses.

[0021] In addition, since the manual pipe bending apparatus of the present disclosure is capable of being manufactured in a small and lightweight type compared to the existing pipe bending apparatuses, the manual pipe bending apparatus of the present disclosure is capable of being carried in a vehicle, so that there is an effect that the bending work of the pipe is capable of being easily performed by bringing the manual pipe bending apparatus of the present disclosure when the bending work of the pipe is required to be performed.

Description

DESCRIPTION OF DRAWINGS

[0022] FIG. **1** is a view illustrating a state in which an operating part of a bending apparatus of the present disclosure is rotated to a rear side up to a first upper connection pin.

[0023] FIG. **2** is a view illustrating an opposite side of the bending apparatus of the present disclosure in the state illustrated in FIG. **1**.

[0024] FIG. **3** is a view illustrating an internal structure of the bending apparatus of the present disclosure in the state illustrated in FIG. **1**.

[0025] FIGS. **4**A, **4**B, **4**C and FIGS. **5**A, **5**B, **5**C show operating states of an operating ratchet and an external gear according to a work process.

DESCRIPTION OF REFERENCE NUMERALS OF MAIN PARTS OF DRAWINGS

[0026] **10**: support **20**: support bracket

[0027] **30**: rotating bracket **30***a*, **30***b*: first and second rotating bracket plates

[0028] **35**: first connection pin

[0029] **40**: bending shoe

[0030] **41**: guide groove **415**: incision part

[0031] **42**: mounting jaw **425**: additional incision part

[0032] **43**: external gear **47**: handle

[0033] **50**: operating part **50***a*, **50***b*: first and second operating plates

[0034] **51**: mounting part **55**: second connection pin

[0035] **60**: operating lever

[0036] **70**: operating ratchet **71**: center part

[0037] **72**: engaging part **73**: catching part

[0038] **80**: elastic means

[0039] S**1**: first rotating shaft S**2**: second rotating shaft

[0040] S**3**: third rotating shaft

MODE FOR INVENTION

[0041] Hereinafter, a manual pipe bending apparatus according to the present disclosure will be described in more detail.

[0042] The manual pipe bending apparatus according to the present disclosure is used by being placed on a floor, and includes a support **10** contacting a ground surface, a support bracket **20** coupled to the support **10**, a rotating bracket **30** coupled to the support bracket **20**, a bending shoe **40** and an operating part **50** that are coupled to the rotating bracket **30**, and an operating lever **60** coupled to the operating part **50** and configured to rotate the operating part **50** up and down.

[0043] The support bracket **20** is configured to support the rotating bracket **30**, and is coupled to a first side of the support **10** such that a side of the support **10** to which the support bracket **20** is not coupled is open, thereby providing a space where a pipe in which bending is finished is capable of

being removed.
[0044] In addition, the support **10** is provided with a guide roller R. The guide roller R is configured to stably support the pipe together with a mounting jaw **42** of the bending shoe **40** which will be described later, and is configured to guide the pipe in a direction toward the bending shoe **40** during a process of bending, the pipe being inserted so as to be bent.
[0045] Next, referring to FIG. **1** to FIG. **3**, in the rotating bracket **30**, the rotating bracket **30** includes first and second rotating bracket plates **30***a* and **30***b* that are connected to each other by a first connection pin **35** with a space formed therebetween.
[0046] That is, the first and second rotating bracket plates **30***a* and **30***b* are in a state in which the first and second rotating bracket plates **30***a* and **30***b* are connected to each other by the first connection pin **35** while the space is formed therebetween. The first connection pin **35** includes a first lower connection pin **35***b* provided on a lower portion of the rotating bracket **30**, and includes a first upper connection pin **35***a* provided on an upper portion of the rotating bracket **30**. Furthermore, in order to maintain a stable coupling state, it is preferable that the first upper connection pin **35***a* and the first lower connection pin **35***b* are positioned diagonally to each other.
[0047] In addition, the first upper connection pin **35***a* is positioned at a rear side of the operating part **50**, and is configured to control a rear rotation angle of the operating part **50**. That is, when the operating part **50** is rotated to the rear side, the first upper connection pin **35***a* prevents the operating part **50** from being further rotated.
[0048] In addition, the bending shoe **40** is coupled such that the bending shoe **40** is capable of being rotated by a first rotating shaft S**1** between the first and second rotating bracket plates **30***a* and **30***b*, and configurations of the bending shoe **40** and an operating ratchet **70** which will be described later have unique technical characteristics of the present disclosure compared to technical characteristics of existing manual pipe bending apparatuses.
[0049] The bending shoe **40** according to the present disclosure has a guide groove **41** to which the inserted pipe is guided as in the existing manual pipe bending apparatuses, but there is a difference that an external gear **43** in addition to the guide groove **41** is formed on an outer peripheral surface of the bending shoe **40**.
[0050] That is, the guide groove **41** is formed on the outer peripheral surface of the bending shoe **40**, but is formed on a part of the outer peripheral surface, and the external gear **43** is formed on the outer peripheral surface where the guide groove **41** is not formed. In addition, since the pipe is bent 15 degrees each time the bending shoe **40** is rotated by pushing the operating lever **60** once downward, the pipe is bent 90 degrees when the same work is performed six times. Therefore, it is preferable that the external gear **43** is designed such that a center angle between a gear and a next gear is 15 degrees.
[0051] Next, the operating part **50** according to the present disclosure is coupled such that the operating part **50** is capable of being rotated in the rotating bracket **30** by a second rotating shaft S**2**. When the operating lever **60** is coupled to the operating part **50** and the operating part **50** is rotated to a front side by a worker pushing the operating lever **60** downward, the operating part **50** serves to rotate the bending shoe **40** by the operating ratchet **70** which will be described later.
[0052] The operating part **50** includes second operating plates **50***a* and **50***b* that are integrally formed by being connected to each other with a second connection pin **55** while a space is formed between the first and second operating plates **50***a* and **50***b*, and the operating ratchet **70** which is configured to be rotated by a third rotating shaft S**3** and which has an end portion engaging with the external gear **43** of the bending shoe **40** is coupled between the first and second operating plates **50***a* and **50***b*.
[0053] Referring to FIG. **3**, the operating ratchet **70** is provided with an elastic force of rotation by an elastic means **80**. Furthermore, the elastic means **80** includes a coil part **81** wound on the third rotating shaft S**3**, a first fixing part **82** connected to and formed on a first end of the coil part **81** and fastened to the operating ratchet **70**, and a second fixing part **83** connected to and formed on a

second end of the coil part **81** and fastened to the second operating plate **50***b*.
[0054] The operating ratchet **70** includes a center part **71** coupled to the third rotating shaft S**3**, an engaging part **72** which is connected to and formed on the center part **71** such that the engaging part **72** is formed toward a first side of the center part **71** and which has an end portion engaging with the external gear **43**, and a catching part **73** which is connected to and formed on the center part **71** such that the catching part **73** is formed toward a second side of the center part **71** and which is configured to control a front rotation of the operating ratchet **70** by the second connection pin **55**. That is, a force causing the operating ratchet **70** to rotate to the front side by the elastic means **80** is controlled since the catching part **73** is caught on the second connection pin **55**.
[0055] Of course, the front rotation of the operating ratchet **70** may be controlled by further adding a configuration other than the second connection pin **55**.
[0056] In addition, a mounting part **51** for mounting the operating lever **60** is provided at an upper portion between the first and second operating plates **50***a* and **50***b*. Furthermore, after the operating lever **60** and the operating part **50** are connected to each other by inserting the operating lever **60** into the mounting part **51**, bending work is performed by pushing the operating lever **60** downward so that the operating lever **60** is rotated to the front side and the bending shoe **40** is also rotated to the front side in conjunction with the operating lever **60**. Furthermore, when the bending work is completed, the operating lever **60** is separated and stored.
[0057] In order for the worker to push the operating lever **60** downward slightly more easily while the operating lever **60** is in a state in which the operating lever **60** is inserted into the mounting part **51**, it is preferable for convenience of the bending work that the mounting part **51** is coupled to the first and second operating plates **50***a* and **50***b* while the mounting part **51** is in a state in which an upper portion of the mounting part **51** is inclined to the front side rather than the mounting part **51** is horizontally coupled between the first and second operating plates **50***a* and **50***b*.
[0058] Next, in the external gear **43** formed on the outer peripheral surface of the bending shoe **40**, inclined surfaces of the gears are not formed at the same inclination, but are formed such that a first gear surface **43**A seen from the front side has a gentle inclination and a second gear surface **43**B has a steep inclination.
[0059] Therefore, in the bending work, pushing work of the operating lever **60** performed by the worker is capable of being easily performed. Furthermore, after bending the pipe once, the operating ratchet **70** is capable of being easily moved to a next neighboring gear for performing the next bending.
[0060] Next, a first side of the guide groove **41** of the bending shoe **40** is provided with the mounting jaw **42** for mounting an end portion of the pipe that is inserted into the bending apparatus along the guide groove **41**, and it is preferable that an incision part **415** in which a part thereof is cut-off is formed on an upper side of the guide groove **41** where the mounting jaw **42** is positioned. This is a configuration that allows the pipe in which the bending work has been completed to be easily removed from the mounting jaw **42**.
[0061] In addition, for the same an additional incision part **425** is formed in a part of the mounting jaw **42** in which a part of an end portion of the mounting jaw **42** is cut-off, so that a space where the pipe in which the bending work has been completed is capable of being easily escaped is provided, thereby facilitating removal of the pipe.
[0062] In addition, a handle **47** is provided on an outer peripheral surface of the mounting jaw **42**, so that the worker can hold the handle **47** and can rotate the bending shoe **40** in a specific direction after starting and completing the bending work.
[0063] An operation process of the bending apparatus according to the present disclosure configured as described above is described as follows.
[0064] First of all, referring to FIGS. **4**A to **4**C, by holding and pushing the handle **47** downward, the bending shoe **40** is rotated to the rear side so that the mounting jaw **42** is positioned at a position where the end portion of the inserted pipe is capable of being mounted thereon.

[0065] In this case, in order to rotate the bending shoe **40** toward the rear side without interference from the operating ratchet **70**, it is preferable that the operating part **50** is also rotated to the rear side. At this time, the rear rotation angle of the operating part **50** is controlled by the first upper connection pin **35***a* (see FIG. **4**A).
[0066] As illustrated in FIG. **4**A, after a preparation is performed by pushing the handle **47** downward such that the pipe is capable of being seated on the mounting jaw **42** of the bending shoe **40**, the worker then inserts the pipe toward the bending shoe **40** along the guide roller R, then the end portion of the pipe is seated on the mounting jaw **42** by being guided along the guide groove **41**, and then a preparation state before the bending work is completed.
[0067] Then, as illustrated in FIG. **4**B, the worker inserts the operating lever **60** to the mounting part **51** of the operating part **50**, and then pushes the operating lever **60** and the operating part **50** downward to a predetermined angle so that a state in which the engaging part **72** of the operating ratchet **70** is engaging with the external gear **43** is realized.
[0068] In this state, as illustrated in FIG. **4**C, when the worker further pushes the operating lever **60** downward, the bending shoe **40** is rotated while the engaging part **72** is in a state in which the engaging part **72** is engaging with the external gear **43**, and bending of the pipe is performed such that the pipe is bent by the rotation angle.
[0069] Then, as illustrated in FIG. **5**A, the worker releases the engaged state of the operating ratchet **70** with respect to the external gear **43** while the worker raises the operating lever **60** and the operating part **50** upward by a predetermined angle. At this time, the engaging part **72** of the operating ratchet **70** is moved to the next external gear **43** by the elastic means **80**. Furthermore, as illustrated in FIG. **5**B, when the operating lever **60** and the operating part **50** is pushed down again, the engaged state of the operating ratchet **70** with respect to the next external gear **40** is realized.
[0070] In addition, as illustrated in FIG. **5**C, as the worker pushes the operating lever **60** and the operating part **50** again, the bending shoe **40** is rotated at the same angle as the described process above while the engaging part **72** of the operating ratchet **70** is the state in which the engaging part **72** is engaging with the next external gear **43**, and the bending of the pipe is performed such that the pipe is bent by the rotation angle.
[0071] The bending of the pipe is performed by repeating the processes described above.
[0072] In this case, when the external gear **43** is formed on the outer peripheral surface of the bending shoe **40** such that the center angle between the gear and the next gear of the external gear **43** is 15 degrees, the pipe is bent 90 degrees while the processes described above are repeated six times.
[0073] Next, when the bending of the pipe is completed to the desired bending angle, the worker separates the operating lever **60** from the operating part **50**, and the worker rotates the operating part **50** up to the first upper connection pin **35***a*, thereby completely releasing the engaged state of the operating ratchet **70** with respect to the external gear **43**.
[0074] In addition, during the bending process, when the worker holds the handle **47** that is gradually raised upward due to the rotation of the bending shoe **40** and then the worker rotates the handle **47** downward (in this case, the state illustrated in FIG. **4**A is realized), the bending shoe **40** is also rotated, and the pipe that is bent is removed from the bending apparatus during this process.
[0075] At this time, due to the incision part **415** formed on the guide groove **41** and the additional incision part **425** formed on the mounting jaw **42**, the bent pipe is capable of being more easily removed while exiting the guide groove **41** and the mounting jaw **42**.
[0076] In describing the present disclosure above, the manual pipe bending apparatus having the specific shape and the specific configuration has been mainly described with reference to the accompanying drawings, but the present disclosure may be variously modified, changed, and substituted by those skilled in the art. Such modification, change, and substitution should be construed as falling within the scope of the present disclosure.

Claims

1. A manual pipe bending apparatus having improved bending work convenience, the manual pipe bending apparatus comprising: a support positioned on a floor; a support bracket coupled to a first side of the support; a rotating bracket coupled to a first side of the support bracket such that the rotating bracket is capable of being rotated, the rotating bracket having first and second rotating bracket plates which are integrally formed by being connected to each other with a space formed therebetween; a bending shoe coupled to the rotating bracket such that the bending shoe is capable of being rotated between the first and second rotating bracket plates by a first rotating shaft, the bending shoe having an outer peripheral surface provided with a guide groove to which a pipe that is inserted is guided; an operating part positioned above the bending shoe and coupled to the rotating bracket such that operating part is capable of being rotated by a second rotating shaft; and an operating lever coupled to a first side of the operating part and configured to control whether the operating part is rotated or not, wherein an external gear is formed on the outer peripheral surface of the bending shoe, and the working part has first and second operating plates that are integrally formed by being connected to each other by a second connection pin with a space formed therebetween, and an operating ratchet which is configured to be rotated by a third rotating shaft and which has an end portion engaging with the external gear of the bending shoe and an elastic means configured to provide an elastic force to the operating ratchet are provided between the first and second operating plates.

2. The manual pipe bending apparatus of claim 1, wherein the external gear is formed such that a first gear surface has a gentle inclination and a second gear surface has a steep inclination.

3. The manual pipe bending apparatus of claim 1, wherein the external gear is formed such that a center angle between one gear and a next gear of the external gear is **15** degrees.

4. The manual pipe bending apparatus of claim 1, wherein the operating ratchet is provided with a center part to which the third rotating shaft is coupled, an engaging part which is connected to and formed on the center part such that the engaging part is formed toward a first side of the center part and which is engaging with the external gear, and a catching part which is connected to and formed on the center part such that the catching part is formed toward a second side of the center part and which is configured to control a front rotation angle of the operating ratchet by the second connection pin.

5. The manual pipe bending apparatus of claim 1, wherein a first connection pin connecting the first and second rotating bracket plates to each other has a first lower connection pin provided on a lower portion of the rotating bracket and a first upper connection pin provided on an upper portion of the rotating bracket, and the first upper connection pin is positioned on a rear side of the operating part and is configured to control a rear rotation angle of the operating part.

6. The manual pipe bending apparatus of claim 1, wherein the bending shoe comprises a mounting jaw which is formed on a first side of the guide groove and on which an end portion of the inserted pipe is mounted, and an incision part in which a part thereof is cut-off is formed on an upper side of the guide groove where the mounting jaw is positioned.

7. The manual pipe bending apparatus of claim 6, wherein an additional incision part in which a part thereof is cut-off is formed on an end portion of the mounting jaw.

8. The manual pipe bending apparatus of claim 6, wherein the mounting jaw is provided with a handle for rotating the bending shoe.

9. The manual pipe bending apparatus of claim 1, wherein the operating lever is capable of being detached from or attached to the operating part.

10. The manual pipe bending apparatus of claim 1, wherein the support is provided with a guide roller configured to stably support the pipe that is inserted so as to be bent.